Energy-Normalized RKHS Pearson Classification: Threshold and Separation Guarantees

1 Pearson score and energy normalization

Let $\mathcal{H}$ be a real separable RKHS with measurable feature map ϕ. For a reference distribution P_i, define

$$\mu_{P_i} = \mathbb{E}_{P_i}\phi(X), \qquad T_{P_i} = \mathbb{E}_{P_i}[\phi(X) \otimes \phi(X)], \qquad A_i = T_{P_i} + \lambda I,$$

where $\lambda > 0$. For an incoming distribution Q, set

$$\Delta_i = \mu_Q - \mu_{P_i}.$$

The regularized RKHS Pearson-type score is

$$D_i = \frac{1}{2}\langle \Delta_i, A_i^{-1}\Delta_i\rangle_{\mathcal{H}}. \tag{1}$$

With

$$v_i = A_i^{-1/2}\mu_Q, \qquad w_i = A_i^{-1/2}\mu_{P_i},$$

this becomes $D_i = \frac{1}{2}\|v_i - w_i\|_{\mathcal{H}}^2$. Thus, A_i^{-1} supplies the reference-dependent geometry.

Define the total embedding energy and normalized score by

$$E_i := \langle \mu_Q, A_i^{-1}\mu_Q\rangle_{\mathcal{H}} + \langle \mu_{P_i}, A_i^{-1}\mu_{P_i}\rangle_{\mathcal{H}}, \tag{2}$$

$$S_i := \frac{D_i}{E_i} = \frac{\|v_i - w_i\|_{\mathcal{H}}^2}{2(\|v_i\|_{\mathcal{H}}^2 + \|w_i\|_{\mathcal{H}}^2)}. \tag{3}$$

When $E_i > 0$, $0 \le S_i \le 1$. This normalization retains the Pearson geometry while measuring relative rather than absolute discrepancy. It defines a normalized score, not the classical Pearson divergence.

2 Sampling assumptions and empirical score

Assumption 1 (Kernel and samples)**.** *The positive-semidefinite kernel is fixed before sampling and satisfies*

$$k(x, y) \ge 0, \qquad k(x, x) \le K$$

for all x, y, with $K > 0$.

The reference and incoming samples satisfy

$$X_1, \ldots, X_n \overset{\text{iid}}{\sim} P_i, \qquad Z_1, \ldots, Z_n \overset{\text{iid}}{\sim} Q,$$

and the two samples are independent.

Define

$$\widehat{\mu}_i = \frac{1}{n}\sum_{j=1}^{n}\phi(X_j), \qquad \widehat{\mu}_Q = \frac{1}{n}\sum_{j=1}^{n}\phi(Z_j),$$

and

$$\widehat{T}_i = \frac{1}{n}\sum_{j=1}^{n}\phi(X_j)\otimes\phi(X_j), \qquad \widehat{A}_i = \widehat{T}_i + \lambda I.$$

The empirical quantities are

$$\widehat{D}_i := \frac{1}{2}\langle \widehat{\mu}_Q - \widehat{\mu}_i, \widehat{A}_i^{-1}(\widehat{\mu}_Q - \widehat{\mu}_i)\rangle_{\mathcal{H}}, \tag{4}$$

$$\widehat{E}_i := \langle \widehat{\mu}_Q, \widehat{A}_i^{-1}\widehat{\mu}_Q\rangle_{\mathcal{H}} + \langle \widehat{\mu}_i, \widehat{A}_i^{-1}\widehat{\mu}_i\rangle_{\mathcal{H}}, \tag{5}$$

$$\widehat{S}_i := \frac{\widehat{D}_i}{\widehat{E}_i}. \tag{6}$$

If $\widehat{E}_i = 0$, define $\widehat{S}_i = 0$. In this case both empirical embeddings, and hence $\widehat{D}_i$, are zero.

The incoming window is accepted as compatible with regime i when $\widehat{S}_i \le \eta$.

Assumption 2 (Reference embedding energy)**.** *A known constant $m_0 > 0$ satisfies*

$$\|\mu_{P_i}\|_{\mathcal{H}} \ge m_0.$$

Assumption 2 prevents the normalization denominator from becoming arbitrarily small under the null. It concerns the reference distribution only. For a finite memory, it may be imposed uniformly over stored regimes or replaced by regime-specific constants.

3 Two useful energy bounds

Introduce the empirical midpoint embedding

$$\widehat{c}_i = \frac{\widehat{\mu}_Q + \widehat{\mu}_i}{2}.$$

The parallelogram identity in the $\widehat{A}_i^{-1}$ metric gives

$$\widehat{E}_i = \widehat{D}_i + 2\langle \widehat{c}_i, \widehat{A}_i^{-1}\widehat{c}_i\rangle_{\mathcal{H}}. \tag{7}$$

Since $\widehat{T}_i \preceq KI$,

$$\langle \widehat{c}_i, \widehat{A}_i^{-1}\widehat{c}_i\rangle_{\mathcal{H}} \ge \frac{\|\widehat{c}_i\|_{\mathcal{H}}^2}{K+\lambda}. \tag{8}$$

We also need an upper bound valid under any incoming distribution.

Lemma 1 (Uniform upper bound on total energy)**.** *Under Assumption 1,*

$$E_i \le M_\lambda, \qquad \widehat{E}_i \le M_\lambda, \qquad M_\lambda := \frac{K}{\lambda} + \frac{K}{K+\lambda}.$$

Proof. The incoming term satisfies

$$\langle \widehat{\mu}_Q, \widehat{A}_i^{-1}\widehat{\mu}_Q\rangle_{\mathcal{H}} \le \frac{\|\widehat{\mu}_Q\|_{\mathcal{H}}^2}{\lambda} \le \frac{K}{\lambda}.$$

The empirical covariance operator is positive semidefinite, so

$$\widehat{T}_i \succeq \widehat{\mu}_i \otimes \widehat{\mu}_i.$$

Inversion reverses the order, giving

$$\langle \widehat{\mu}_i, \widehat{A}_i^{-1}\widehat{\mu}_i\rangle_{\mathcal{H}} \le \frac{\|\widehat{\mu}_i\|_{\mathcal{H}}^2}{\lambda + \|\widehat{\mu}_i\|_{\mathcal{H}}^2} \le \frac{K}{K+\lambda}.$$

The population proof is identical, using $T_{P_i} \succeq \mu_{P_i} \otimes \mu_{P_i}$. □

4 Null guarantee

For $\rho_0 \in (0,1)$, define

$$U_0 := \frac{K}{n\lambda}\left(1+\sqrt{\log\frac{2}{\rho_0}}\right)^2, \tag{9}$$

$$a_0 := \sqrt{\frac{K}{2n}}\left(1+\sqrt{\log\frac{2}{\rho_0}}\right), \tag{10}$$

$$H_0 := \frac{2[m_0-a_0]_+^2}{K+\lambda}, \tag{11}$$

$$\eta_0 := \frac{U_0}{U_0+H_0}. \tag{12}$$

Theorem 1 (Acceptance under the null)**.** *Under Assumptions 1 and 2, if $Q = P_i$, then*

$$\mathbb{P}\{\widehat{S}_i \le \eta_0\} \ge 1-\rho_0.$$

The threshold is strictly smaller than one whenever $m_0 > a_0$.

Proof. Under the null,

$$\mathbb{E}\|\widehat{\mu}_Q - \widehat{\mu}_i\|_{\mathcal{H}}^2 \le \frac{2K}{n}.$$

Nonnegativity of the kernel implies

$$\|\phi(x)-\phi(y)\|_{\mathcal{H}} \le \sqrt{2K}.$$

Applying one-sided McDiarmid concentration to the norm of the mean difference, and using $\widehat{A}_i^{-1} \preceq \lambda^{-1} I$, gives

$$\widehat{D}_i \le U_0$$

with probability at least $1-\rho_0/2$.

Under $Q = P_i$, the midpoint $\widehat{c}_i$ is the empirical mean of $2n$ iid observations from P_i. Its expected norm error is at most $\sqrt{K/(2n)}$. The same bounded-difference argument gives

$$\|\widehat{c}_i - \mu_{P_i}\|_{\mathcal{H}} \le a_0$$

with probability at least $1-\rho_0/2$.

On the intersection of these events,

$$\|\widehat{c}_i\|_{\mathcal{H}} \ge [m_0-a_0]_+.$$

Equations (7) and (8) therefore imply

$$\widehat{E}_i \ge \widehat{D}_i + H_0.$$

When $H_0 > 0$,

$$\widehat{S}_i \le \frac{\widehat{D}_i}{\widehat{D}_i+H_0} \le \frac{U_0}{U_0+H_0}.$$

If $H_0 = 0$, the claim follows from $\widehat{S}_i \le 1 = \eta_0$. A union bound completes the proof. □

5 Alternative guarantee from Pearson separation

Assumption 3 (Pearson separation)**.** *For the fixed pair (Q, P_i),*

$$D_i \geq \delta > 0.$$

For $\rho_1 \in (0,1)$, define

$$s_1 = \sqrt{\frac{K \log(2/\rho_1)}{n\lambda}}, \qquad r_1 = \frac{K}{\lambda}\sqrt{\frac{\log(2/\rho_1)}{2n}}. \tag{13}$$

Theorem 2 (Rejection of a separated regime)**.** *Under Assumptions 1 and 3,*

$$\mathbb{P}\left\{\widehat{S}_i \geq L_S(\delta, \rho_1)\right\} \geq 1 - \rho_1,$$

where

$$L_S(\delta, \rho_1) := \frac{[\sqrt{\delta} - s_1]_+^2}{M_\lambda(1 + r_1)}. \tag{14}$$

Consequently, if $\eta < L_S(\delta, \rho_1)$, then

$$\mathbb{P}\{\widehat{S}_i > \eta\} \geq 1 - \rho_1.$$

This alternative guarantee does not require Assumption 2.

Proof. Define the fixed population direction

$$u_i = \frac{A_i^{-1}\Delta_i}{\sqrt{2D_i}}.$$

It satisfies

$$\langle u_i, \Delta_i\rangle_{\mathcal{H}} = \sqrt{2D_i}, \qquad \langle u_i, A_i u_i\rangle_{\mathcal{H}} = 1, \qquad \|u_i\|_{\mathcal{H}}^2 \leq \lambda^{-1}.$$

Set

$$\widehat{d}_i = \langle u_i, \widehat{\mu}_Q - \widehat{\mu}_i\rangle_{\mathcal{H}}, \qquad \widehat{q}_i = \langle u_i, \widehat{A}_i u_i\rangle_{\mathcal{H}}.$$

The variational representation of the empirical Pearson score, restricted to nonnegative multiples of u_i, gives

$$\widehat{D}_i \geq \sup_{t \geq 0}\left\{t\widehat{d}_i - \frac{t^2}{2}\widehat{q}_i\right\} = \frac{[\widehat{d}_i]_+^2}{2\widehat{q}_i}.$$

For the directional mean,

$$|u_i(x) - u_i(y)| \leq \sqrt{2K}\|u_i\|_{\mathcal{H}}.$$

Hoeffding concentration for the $2n$ independent summands yields

$$\widehat{d}_i \geq \sqrt{2D_i} - \sqrt{2}\, s_1$$

with probability at least $1 - \rho_1/2$.

For the quadratic term,

$$0 \leq u_i(x)^2 \leq K\|u_i\|_{\mathcal{H}}^2, \qquad \mathbb{E}\widehat{q}_i = 1.$$

A second application of Hoeffding gives

$$\widehat{q}_i \leq 1 + r_1$$

with probability at least $1 - \rho_1/2$.

By a union bound, with probability at least $1 - \rho_1$,

$$\widehat{D}_i \geq \frac{[\sqrt{D_i} - s_1]_+^2}{1 + r_1} \geq \frac{[\sqrt{\delta} - s_1]_+^2}{1 + r_1}.$$

Lemma 1 gives $\widehat{E}_i \leq M_\lambda$, so dividing proves the claim. If $\widehat{E}_i = 0$, then $\widehat{D}_i = 0$; on the event above this is possible only when its stated lower bound is zero, in which case the result still holds. □

6 Nonempty threshold interval

The null and alternative guarantees are compatible when

$$\eta_0 \le \eta < L_S(\delta, \rho_1). \tag{15}$$

Since $\eta_0 > 0$, this certified interval is nonempty if and only if

$$\boxed{\delta > \left[s_1 + \sqrt{M_\lambda(1+r_1)\eta_0}\right]^2.} \tag{16}$$

This equivalence concerns the displayed bounds, not a necessary condition for the classifier to succeed.

Taking $\rho_0 = \rho_1 = \rho$, set

$$\ell = \log\frac{2}{\rho}, \qquad U = \frac{K}{n\lambda}(1+\sqrt{\ell})^2, \qquad H = \frac{2}{K+\lambda}\left[m_0 - \sqrt{\frac{K}{2n}}(1+\sqrt{\ell})\right]_+^2.$$

A valid threshold is

$$\boxed{\eta = \frac{U}{U+H}.} \tag{17}$$

The corresponding sufficient Pearson separation is

$$\boxed{\delta > \left[\sqrt{\frac{K\ell}{n\lambda}} + \sqrt{\left(\frac{K}{\lambda} + \frac{K}{K+\lambda}\right)\left(1 + \frac{K}{\lambda}\sqrt{\frac{\ell}{2n}}\right)\frac{U}{U+H}}\right]^2.} \tag{18}$$

A useful null threshold requires

$$m_0 > \sqrt{\frac{K}{2n}}\left(1 + \sqrt{\log\frac{2}{\rho}}\right).$$

For fixed positive m_0, K, λ, and ρ, the threshold and the sufficient separation in (18) are both of order n^{-1}.

7 Interpretation and limitations

Additional requirement introduced by normalization. Minimum Pearson separation controls the alternative numerator. It does not prevent the null denominator from being small. The reference embedding lower bound m_0 addresses this separate issue.

For independent $X, X' \sim P_i$,

$$\|\mu_{P_i}\|_{\mathcal{H}}^2 = \mathbb{E}[k(X, X')].$$

Thus, m_0 describes within-regime kernel similarity. The diagonal normalization $k(x, x) = 1$ alone does not supply a useful distribution-uniform positive value of m_0.

Conservatism of this study. The alternative proof combines a directional Pearson lower bound with a uniform upper bound on total energy. It is a valid sufficient guarantee, but it does not establish that normalization improves the separation requirement relative to unnormalized Pearson. A sharper analysis would exploit the joint structure of the numerator and denominator.

Pairwise versus complete classification. The theorems concern one fixed reference comparison. For a memory of several regimes, correct assignment requires simultaneous acceptance of the correct regime and rejection of incorrect regimes. Appropriate error allocation is needed to combine these events. Repeated monitoring, data-dependent bandwidths, and windows selected after detection also require additional justification beyond the fixed-sample assumptions used here.